Audio Deepfake Detection

Using Utterance-Level Embeddings

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About the Project

Objective

Detect AI-generated (deepfake) audio and distinguish it from genuine human speech.

Problem Statement

- Binary classification: **Real** vs. **Fake** audio
- Addresses the threat of AI voice spoofing
- Operates on pre-extracted utterance-level embeddings

Real-World Applications

- Fraud prevention
- Security systems
- Media verification
- Voice authentication

Dataset & Features

Data Source

Pre-extracted utterance-level embeddings from the **WavLM** pretrained speech model, stored as `.npy` files.

- **Feature dimensions:** 768 per utterance
- Each embedding is a compact numerical representation of one audio clip
- **Class 0:** Real (Bonafide) speech
- **Class 1:** Fake (Spoofed) speech

Data Format

$$X = [\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n]$$

$$\mathbf{e}_i \in \mathbb{R}^{768}$$

$$Y = [0, 1, 0, 1, \ldots]$$

Data Preprocessing

Pipeline Steps

1. Load `.npy` embedding files & assign labels
2. Standardize features using `StandardScaler`
3. Handle class imbalance
4. Split into train / validation / test sets

Class Imbalance Strategies

- **Downsampling (DS):** Reduce majority class
- **SMOTE:** Synthetic Minority Over-sampling

t-SNE Visualization

The t-SNE plot reveals partial overlap between real (red) and spoofed (blue) embedding clusters, motivating non-linear classifiers for best separation.

Models Overview

Classical ML

- Support Vector Machine (SVM)
- Decision Tree
- Random Forest
- k-Nearest Neighbours (KNN)
- Gaussian Bayes (custom)
- Gaussian Naive Bayes
- Logistic Regression
- Perceptron

Deep Learning

- MLP — Single hidden layer
- MLP — Double hidden layers

Evaluation Metrics

- Accuracy, Precision, Recall
- F1-Score (harmonic mean)
- Confusion Matrix

Neural Networks

Best Architecture: Double Hidden Layer

- Input(768) $\rightarrow$ H1(768) $\rightarrow$ H2(256) $\rightarrow$ Output(1)
- Activation: **ReLU** (hidden), **Sigmoid** (output)
- Optimizer: **Adam** (lr $= 0.01$)
- Loss: Binary Cross-Entropy
- Early stopping: patience $=$ 10–15 epochs
- Best weights saved via `copy.deepcopy`

Test Performance

Metric	Score
Accuracy	98.74%
Real Precision	99.93%
Real Recall	97.55%
Real F1	98.73%
Fake Precision	97.61%
Fake Recall	99.93%
Fake F1	98.76%

Support Vector Machine (Best Overall)

Architecture & Tuning

- Kernel: **RBF** with `gamma='scale'`
- Regularization search: $C \in \{0.1, 1.0, 10.0\}$
- **Best config:** SVM with $C = 10.0$ on SMOTE dataset
- Emerged as **strongest model** in final evaluation

Key Insight

SMOTE balancing + high C value captured complex non-linear boundaries in the 768-dimensional embedding space.

Test Performance

Metric	Score
Accuracy	**99.16%**
Real Precision	100.00%
Real Recall	98.32%
Real F1	99.15%
Fake Precision	98.34%
Fake Recall	100.00%
Fake F1	99.16%

k-NN and XGBoost

k-Nearest Neighbours

Best: $k = 5$, Downsampled dataset

- Distance-based majority vote
- $k \in \{3, 5, 7, 11\}$ tested

Metric	Score
Accuracy	95.53%
Real F1	95.36%
Fake F1	95.69%

XGBoost

Best: 200 trees, depth=3, lr=0.3, SMOTE

- Sequential decision tree ensemble
- Grid: estimators, depth, learning rate

Metric	Score
Accuracy	97.91%
Real F1	97.87%
Fake F1	97.95%

Decision Tree, Random Forest & Gaussian Bayes

Decision Tree

Max depth: 10 (regularised)

Accuracy	94.15%
Real F1	94.00%
Fake F1	94.00%

Interpretable baseline for ensemble comparison.

Random Forest

100 trees, depth=15, `n_jobs=-1`

Accuracy	96.24%
Real F1	96.00%
Fake F1	96.00%

80/20 stratified split.

Gaussian Bayes

Custom multivariate covariance model outperforms standard Naive Bayes.

Accuracy	89.90%
Real F1	88.76%
Fake F1	90.83%

Perceptron & Logistic Regression

Perceptron

`max_iter=1000`, `random_state=42`

Metric	Score
Accuracy	99.04%
Real Precision	99.70%
Real Recall	98.38%
Fake Precision	98.40%
Fake Recall	99.70%

Logistic Regression

`max_iter=1000`, `StandardScaler` fitted on train only

Metric	Score
Accuracy	99.02%
Real Precision	99.70%
Real Recall	98.38%
Fake Precision	99.70%
Fake Recall	98.38%

Both linear models performed surprisingly well, confirming the WavLM embeddings are nearly linearly separable.

Model Comparison — Accuracy

Model	Best Config	Accuracy
accentblue!20 SVM	RBF, C=10, SMOTE	**99.16%**
Perceptron	Default, full data	99.04%
Logistic Reg.	L_2, max_iter=1000	99.02%
Neural Network	768→256, Adam	98.74%
XGBoost	n=200, depth=3, SMOTE	97.91%
Random Forest	100 trees, depth=15	96.24%
k-NN	k=5, Downsampled	95.53%
Decision Tree	depth=10	94.15%
Gaussian Bayes	Custom multivariate	89.90%
Naive G. Bayes	Standard sklearn	≈88%

Model Comparison — Precision, Recall & F1

Precision Rankings (approx.)

- SVM variants & k-NN SMOTE: ≈**100%**
- XGBoost, SVM Linear: ≈99.9%
- Logistic Reg., MLP: ≈99.7%
- Random Forest: ≈100%
- Gaussian Bayes: ≈91%
- Naive Gaussian Bayes: ≈90%

Recall Rankings (Fake class)

- SVM C=10 SMOTE: **100.0%**
- XGBoost variants: 99.90%
- SVM Poly, Linear: ≈100%
- Logistic Reg. / MLP: ≈99.7%
- k-NN SMOTE: ≈99.1%
- Random Forest: ≈100%
- Gaussian Bayes: ≈90% (lowest)

SVM (RBF, C=10, SMOTE) leads across all four metrics.

Confusion Matrix Summary — Top 3 Models

SVM (Best)

	Pred R	Pred F
Real	5876	124
Fake	1	5999

Only 1 fake misclassified!

Perceptron

	Pred R	Pred F
Real	5903	97
Fake	18	5982

Very clean separation.

Neural Network

	Pred R	Pred F
Real	5849	151
Fake	6	5994

Only 6 fakes missed.

All top models show highly concentrated diagonal confusion matrices.

Decision Boundaries (2D PCA Projection)

Linear models (Logistic Regression, Perceptron)

- Single straight hyperplane separates classes
- Surprisingly effective due to rich WavLM embeddings

Neural Network (Double Hidden)

- Smooth curved boundary
- Captures non-linear structure

SVM (RBF)

- Curved, smooth non-linear boundary
- Best separation in PCA projection

Decision Tree / Random Forest

- Axis-aligned rectangular splits
- More irregular boundaries in PCA space

2D PCA projections are for visualization only; all models train on 768-dimensional embeddings.

Impact of Class Balancing

SMOTE (Over-sampling)

- Generates synthetic minority samples
- Helped SVM and XGBoost most
- SVM C=10 on SMOTE ⇒ **best overall model**
- XGBoost on SMOTE: 97.91%

Downsampling

- Strips noisy majority samples
- Helped KNN generalise better (k=5, DS)
- XGBoost DS also competitive at 97.91%
- Reduces training time significantly

Key Takeaway

The best balancing strategy depends on the model: SMOTE benefits margin-based and boosting methods; downsampling benefits instance-based methods like k-NN.

Why WavLM Embeddings Work So Well

- **WavLM** is a self-supervised speech foundation model trained on large-scale audio data
- 768-dimensional utterance-level embeddings encode deep acoustic and linguistic patterns
- Genuine and synthetic voices leave **different spectral/temporal fingerprints** captured in the embedding space
- High-quality features enable even simple models (Perceptron, Logistic Regression) to achieve $>99\%$ accuracy
- Non-linear models (SVM RBF, NN) exploit residual non-linearity for the final precision gains

t-SNE Cluster Analysis

- Real class (red): multiple tight sub-clusters
- Fake class (blue): large diffuse region
- Partial overlap zone explains the $\sim 1\%$ error rate across all models
- No model perfectly separates the overlap region

Summary of Results

Top 3 Models

1. **SVM (RBF, C=10, SMOTE)**
   - Accuracy: 99.16% — Fake Recall: 100%
2. **Logistic Regression**
   - Accuracy: 99.02% — Fake Recall: 98.38%
3. **Neural Network (768→256)**
   - Accuracy: 98.74% — Fake Recall: 99.93%

Key Findings

- WavLM embeddings are highly discriminative for deepfake detection
- SVM with SMOTE achieves near-perfect fake recall (zero missed fakes on test set)
- Linear models perform surprisingly strongly, confirming good feature quality
- Tree-based models lag slightly due to axis-aligned decision boundaries

Conclusion & Future Work

Conclusions

- Successfully built a binary audio deepfake detection system
- Evaluated 10+ model configurations across 7 algorithm families
- SVM (RBF, C=10) on SMOTE-balanced WavLM features is the recommended production model
- The 768-dim embeddings are the critical enabler of high accuracy

Future Directions

- Fine-tune WavLM end-to-end for deepfake detection
- Test on cross-dataset generalization (unseen TTS systems)
- Explore ensemble/stacking of top models
- Real-time streaming deepfake detection pipeline
- Investigate explainability (SHAP/LIME) on embedding dimensions

Thank You

Audio Deepfake Detection using Utterance-Level Embeddings

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